

From Single-Asset Timing to Portfolio Allocation

A Three-Phase Exploration of Backtesting Engines, Volatility Controls, and Modern Portfolio Theory

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GitHub Repository: <https://github.com/ricardo05143018-creator/quant-backtest-engine>

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Abstract

This project documents the empirical development of a custom Python backtesting engine evaluated across three distinct phases of quantitative strategy design. In Phase 1, a bar-by-bar simulation loop restricted to $t - 1$ closure data was constructed to eliminate look-ahead bias. Testing indicated that static risk boundaries, specifically a fixed 5% stop-loss, induce severe whipsaw degradation on high-beta assets such as TSLA and BTC-USD due to serial dependencies in volatility regimes. Phase 2 implemented a volatility-adaptive Average True Range (ATR) mechanism, validating its single-asset efficiency while highlighting structural calibration sensitivity during cross-asset deployment.

To mitigate the instability of single-asset parameter tuning, Phase 3 transitioned the research framework to cross-asset allocation via Markowitz Modern Portfolio Theory (MPT), integrating Gold (GLD) as a low-correlation defensive asset. The numerical outputs of the Scipy mathematical optimizer were audited and verified using a 10,000-portfolio brute-force Monte Carlo simulation to map the empirical Efficient Frontier. Finally, an out-of-sample (OOS) stress test extending into mid-June 2026 demonstrated that while mean-variance optimization exhibits high sensitivity to parameter drift during macro regime shifts—with the Max Sharpe portfolio’s absolute metrics decaying to an OOS Sharpe ratio of -0.07 (-2.53% cumulative return)—the underlying historical covariance structure provided robust structural downside insulation compared to a naive $1/N$ benchmark, which plummeted to a Sharpe ratio of -0.66 (-6.88% cumulative return).

1 Introduction and Motivation

A systematic review of entry-level quantitative trading tutorials indicates a reliance on over-simplified vectorized backtests. Performing execution logic across an entire pandas dataset simultaneously introduces severe data leakage and look-ahead bias. In an operational trading environment, path-dependent decisions must be conditioned strictly on historical information.

The primary objective of this project was to design a robust, event-simulating execution pipeline that evolves through three algorithmic iterations:

1. **Phase 1 (Engineering & Fixed Risk):** Engineering a strict day-by-day execution loop to eliminate future data leakage, and analyzing the structural breakdown of a fixed 5% stop-loss threshold under heterogeneous volatility regimes.
2. **Phase 2 (Adaptive Risk & Overfitting):** Implementing a dynamic, volatility-adjusted ATR (Average True Range) stop-loss cushion and testing parameter robustness via cross-asset validation.
3. **Phase 3 (Asset Allocation):** Eliminating isolated asset parameter optimization in favor of Modern Portfolio Theory (MPT) to reduce portfolio variance mathematically via covariance minimization.

The progression of this engine demonstrates that resolving basic data leakage constraints often uncovers deeper statistical anomalies. Isolating and identifying the mathematical causes of

model failure under real-world market constraints represents the core analytical value of this research.

2 Phase 1: Base Engine Architecture & The Fixed Stop-Loss Flaw

2.1 Eliminating Look-Ahead Bias

To prevent forward information leakage, the engine establishes a strict engineering constraint: **any trading execution on day t can only evaluate vectors generated at or prior to the close of day $t - 1$** . The engine utilizes a close-to-close execution model where trading signals are calculated using the terminal parameters of day $t - 1$, executing market orders near closes[$t-1$] to position capital to capture the return of day t . While this iterative architecture increases computational overhead relative to vectorized pandas operations, it remains a logical necessity to mirror live execution metrics accurately.

2.2 The Whipsaw Effect on Volatile Assets

A Dual Moving Average (DMA) crossover model with a static 5% stop-loss was evaluated across AAPL, TSLA, and BTC-USD using 2025 market data. A baseline grid search on AAPL identified MA(8, 60) as the optimal parameter allocation. On the low-volatility profile of AAPL, the strategy executed a single entry, maintained the long position for 126 trading days without breaching the threshold, and yielded a +31.6% return (Sharpe 2.16).

However, executing identical parameters on higher-beta assets revealed significant structural vulnerabilities. On TSLA, the engine executed four distinct entries with a mean holding period of only 17 days per trade, finishing the backtest at -5.4% while the underlying buy-and-hold asset appreciated by approximately 80%. On BTC-USD, two of the three initiated positions triggered immediate stop-outs within minimal trading windows.

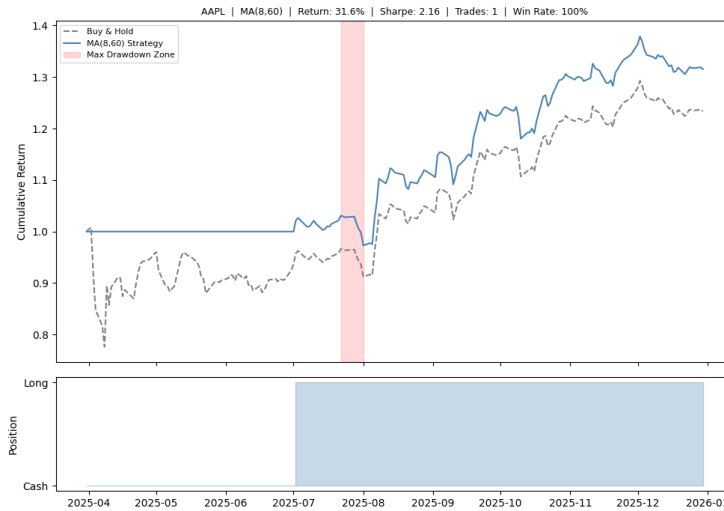


Figure 1. Phase 1: Benchmark performance on AAPL (2025). The low-volatility asset cleanly tracks the trend-following strategy without hitting the fixed 5% threshold.

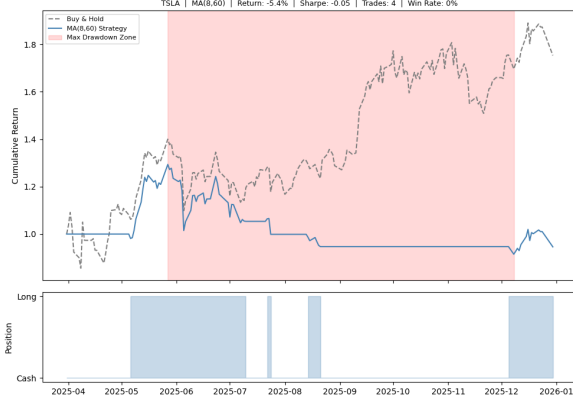


Figure 2. Phase 1: Whipsaw on TSLA.

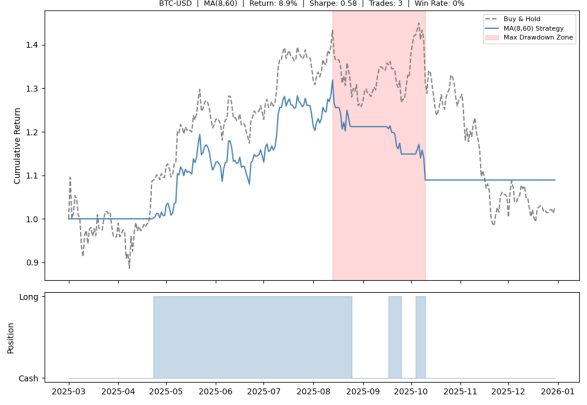


Figure 3. Phase 1: Whipsaw on BTC-USD.

As mapped in Figure 2 and Figure 3, the statistical mechanism driving this decay is the serialization of false exits. High intraday and daily variance on TSLA and BTC regularly breach the rigid 5% boundary during broader upward trends. The engine executes early liquidation near local price minima, subsequently re-entering at higher cost bases upon the next trend verification. This cyclical behavior systematically erodes equity capital. The static stop-loss model assumes homoscedastic volatility across distinct asset classes, an assumption which is empirically invalid.

3 Phase 2: ATR Controls and The Overfitting Trap

To normalize risk across varying volatility profiles, a rolling 14-day Average True Range (ATR) metric was integrated to account for structural price jumps and overnight gaps. The liquidation logic was updated to execute only when price action violated a dynamic trailing cushion defined as: $\text{Entry Price} - (\text{Multiplier} \times \text{ATR})$.

A 3D grid search executed on the AAPL dataset identified $1.5 \times \text{ATR}$ as the optimized local threshold, effectively insulating the strategy from false liquidations during mid-year horizontal consolidations (Figure 4).

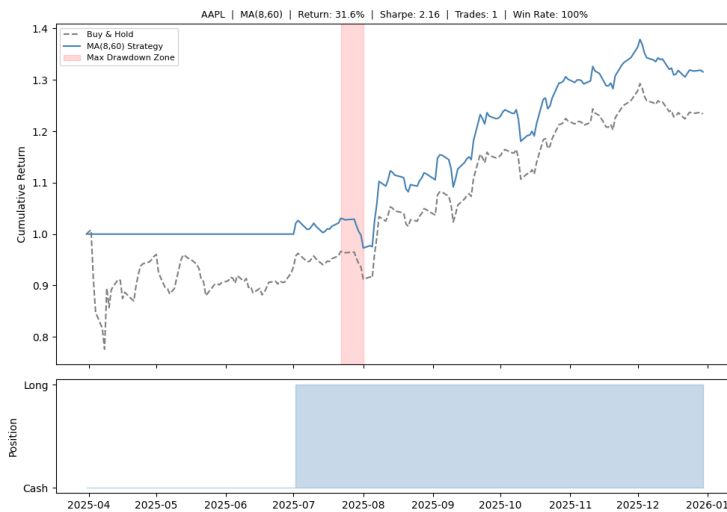


Figure 4. Phase 2: Optimised ATR stop-loss on AAPL (2025), yielding cleaner trend preservation.

3.1 Cross-Asset Calibration Sensitivity

When this specific $1.5 \times \text{ATR}$ parameter allocation was subjected to a cross-asset robustness audit on TSLA and BTC-USD without re-calibration, the performance metrics decayed significantly (Figure 5 and Figure 6).

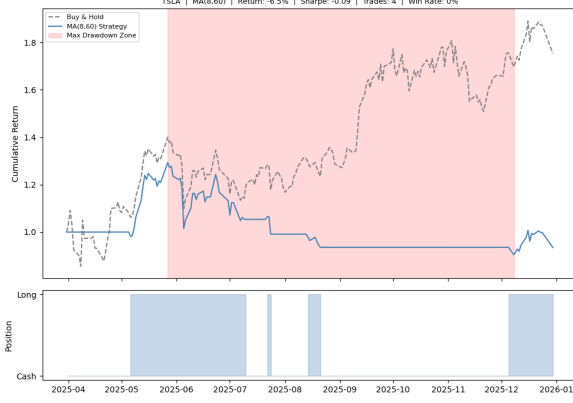


Figure 5. Phase 2: ATR failure on TSLA.

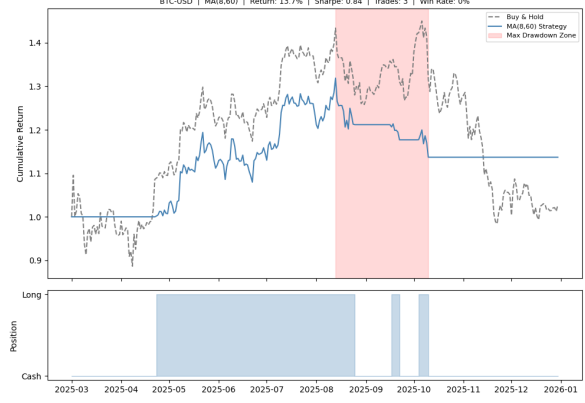


Figure 6. Phase 2: ATR failure on BTC.

This deterioration highlights the limitations of historical curve-fitting. The empirical grid search algorithm over-fit the low-volatility covariance matrix specific to AAPL in 2025. When exposed to the high-beta profile of TSLA or the discrete macro price gaps of Bitcoin, the narrow parameter triggered continuous false stops near short-term cycle lows. This indicates that tuning path-dependent risk parameters on isolated historical asset vectors yields models highly sensitive to regime shifts.

4 Phase 3: Modern Portfolio Theory (MPT) & Covariance

Phase 2 confirmed that optimizing directional risk parameters on isolated equities fails under changing market conditions. To minimize systemic variance, the engine architecture was transitioned from single-asset market timing to multi-asset allocation. SPDR Gold Shares (GLD) was introduced into the investment pool as a low-correlation asset class. To resolve the engineering challenges of frequent remote API calls and data-throttling under execution, a local file-caching layer (`v3_market_data_2025.csv`) was developed. The engine extracted a daily log-return matrix to compute the 2025 asset correlation coefficients (Figure 7).

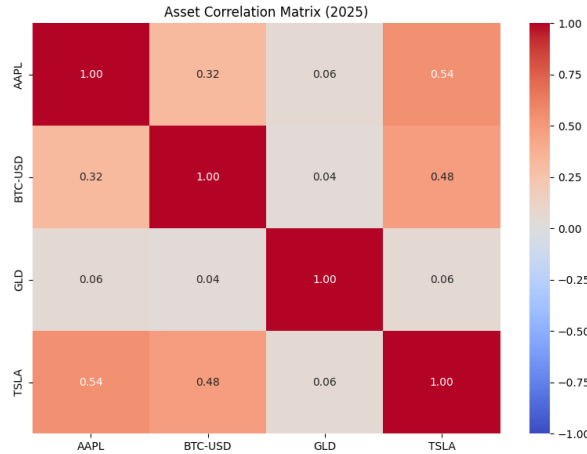


Figure 7. Phase 3: 2025 Asset Correlation Matrix. Note that GLD exhibits near-zero correlation with tech equity and crypto pools.

Because GLD exhibits near-zero correlation coefficients relative to tech equities and crypto assets ($0.035 \sim 0.057$), its inclusion minimizes total portfolio variance (σ_p^2) by reducing the cross-covariance terms within the Markowitz formulation.

The `scipy.optimize` non-linear solver was deployed to resolve two target allocation weights based on 2025 parameters:

- **Global Minimum Variance (GMV):** Allocated 72.9% to GLD, 16.0% to AAPL, 11.1% to BTC, and 0.0% to TSLA (Expected Return = 23.46%, Volatility = 15.50%, Sharpe = 1.51).
- **Maximum Sharpe Ratio:** Allocated 87.1% to GLD, 12.9% to AAPL, and 0.0% to the remaining assets (Expected Return = 28.14%, Volatility = 16.34%, Sharpe = 1.72).

5 Brute-Force Verification via Monte Carlo Simulation

While the formal multi-variable convergence mechanics of the SciPy SLSQP solver utilize calculus-based optimization algorithms outside the scope of this study, its empirical outputs were audited numerically. Treating the numerical solver as a standard functional optimization component, the engine was programmed to execute a brute-force Monte Carlo simulation, randomly generating 10,000 distinct portfolio weight combinations under the constraint that $\sum w_i = 1$.

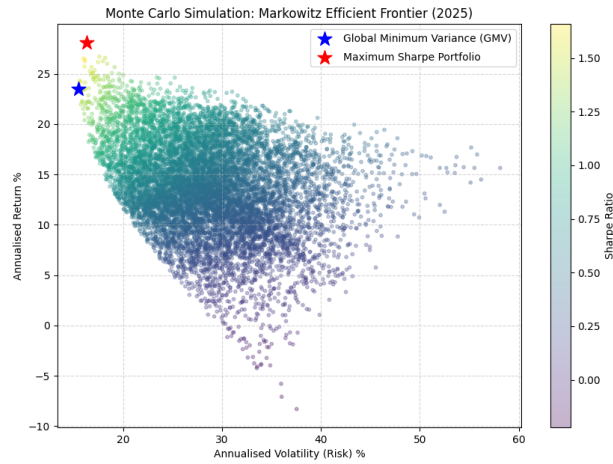


Figure 8. Monte Carlo Simulation of 10,000 random portfolios. The mathematical optimal solutions provided by Scipy perfectly land on the boundaries of the brute-force point cloud.

As shown in Figure 8, the point cloud constructed via random generation maps the classic parabolic Markowitz Efficient Frontier. The SciPy optimizer’s GMV allocation (Blue Star) and Max Sharpe allocation (Red Star) land exactly on the upper efficiency boundary, visually demonstrating the convergence properties of the numerical solver.

5.1 Out-of-Sample Validation and Optimization Instability

The allocation metrics in Section 4 expose an operational vulnerability inherent to mean-variance optimization. The 2025 Max Sharpe configuration concentrated a massive 87.1% of portfolio capital into Gold, heavily weighting the asset due to its stable trend profile throughout the training period.

To evaluate the robustness of this allocation under a shifting macroeconomic environment, an out-of-sample (OOS) stress test was conducted by freezing the optimized 2025 weights and

running them through unseen market data from January 1 to June 15, 2026. The empirical results indicate significant parameter sensitivity: the in-sample Sharpe ratio of 1.72 collapsed completely under the new regime, yielding a negative OOS Sharpe ratio of -0.07 and an absolute cumulative return of -2.53% .

Despite generating negative absolute performance during this highly volatile window, a comparative analysis against a naive $1/N$ equal-weight benchmark reveals an important structural insight. The $1/N$ benchmark experienced a severe liquidation draw-down, plunging to an OOS Sharpe ratio of -0.66 and a cumulative return of -6.88% . The equal-weight model experienced substantial decay because it mechanically forced a combined 50% capital allocation into high-beta vectors (TSLA and BTC-USD) during an aggressive market down-cycle.

Conversely, the optimized Max Sharpe portfolio mitigated this downside risk. This protection was not driven by accurate forward alpha prediction, but was an advantage stemming from its 87.1% concentration in Gold. Gold's low correlation structure insulated the total capital base from the severe tech and crypto sell-offs. This illustrates that while mean-variance models often act as *error maximizers* regarding expected returns by over-fitting to historical trends, the asset covariance structures they map can still provide effective out-of-sample risk isolation during market liquidations.

6 Conclusion and Academic Aspirations

The iterative development from Phase 1 to Phase 3 indicates that quantitative strategy design requires rigorous statistical validation rather than simple coding fluency. Path-dependent simulation loops are required to prevent look-ahead biases; isolated parameter grid searches lead to fragile curve-fitting; and portfolio variance reduction is achieved more reliably through systematic asset covariance mapping than through localized market timing. Finally, out-of-sample testing indicates that optimization frameworks that treat historical returns as static forward expectations remain vulnerable to macro regime shifts.

Deploying computational brute-force to verify optimization boundaries provided an objective method to evaluate advanced numerical tools. These quantitative insights directly motivate my intention to pursue formal undergraduate studies in mathematics and statistics. Acquiring advanced linear algebra and multivariable calculus tools is necessary to develop and audit mathematical models capable of handling market non-stationarity and statistical uncertainty rigorously.